Astrophysics/Physics Honours Research Skills Class 5: How to use AI tools

March 2026

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1. AI Basics

1.1 What AI Actually Is (and Isn't)

"AI" in the context of your daily tools means **large language models (LLMs)** — deep learning systems trained on vast text corpora to predict and generate language. They are not thinking machines; they are sophisticated pattern-completion engines.

Why this matters for you:

- LLMs have **no ground-truth model of physics**. They will confidently produce nonsense if unchecked.
- They excel at synthesis, format translation, and generating plausible text — not at reasoning from first principles.
- Astronomy also relies on **classical ML** (random forests, CNNs, Bayesian networks) for data pipelines — these are fundamentally different tools from the chatbots you interact with.

> **Astronomy example:** Oxford researchers used Google's Gemini with just 15 example images and plain-language instructions to classify real cosmic transients vs. imaging artefacts at ~93% accuracy (Stoppa & Bulmus et al., *Nature Astronomy*, 2025). This is an LLM doing pattern recognition, not astrophysics.

Ref: Djorgovski et al., arXiv:2212.01493 (2022); Stoppa & Bulmus et al., Nature Astronomy (2025).

1.2 The Provider Landscape

No single provider dominates. You should be comfortable switching between them.

Provider	Current Models (March 2026)	Strengths
OpenAI	GPT-5.4 (Thinking/Pro), GPT-5.3 Codex	Largest ecosystem, strong agentic coding, versatile tooling
Anthropic	Claude Opus 4.6, Claude Sonnet 4.6	Leading in coding/agentic work, long context (200K), careful reasoning
Google	Gemini 3.1 Pro, 3.1 Flash-Lite	Google ecosystem integration, strong multimodal, competitive pricing
Meta	Llama 4 (Scout, Maverick)	Open-weight, run locally — critical for data privacy
DeepSeek	DeepSeek V4	Open-source competitor, strong reasoning, very cost-effective

Model names change fast — the table above will date quickly. The point is the ecosystem: OpenAI and Anthropic lead the proprietary frontier, Google integrates tightly with Workspace, and open-weight models (Meta, DeepSeek) let you run locally when privacy matters.

1.3 Access Tiers: Free, Paid, and API

Three ways to use LLMs, with very different capabilities:

- **Free tiers** (ChatGPT free, Claude free, Gemini) — rate-limited models with restricted features. Fine for casual queries, insufficient for research.
- **Budget tiers** (ChatGPT Go at $8/month) — a middle ground with access to current models but limited reasoning and multimedia features.
- **Paid subscriptions** (~$20/month for ChatGPT Plus, Claude Pro, Google AI Pro) — latest models, longer context, file uploads, deep research, extended thinking. **A paid subscription to at least one provider is a worthwhile research investment.**

- **API access** (pay-per-token) — programmatic control for batch processing, pipeline integration, or custom tools. Becomes essential when you need to process data at scale.

> **Practical tip:** Prototype a prompt in the chat interface, then deploy it via the API when you need to run it over hundreds of files.

Ref: Provider pricing pages (as of March 2026); Wiley AI Guidelines for Researchers (2025).

1.4 Apps, SDKs, and Agentic IDEs

There are now three distinct interfaces for working with LLMs:

- **Chat apps** (web/mobile) — best for interactive work: brainstorming, drafting, debugging, literature questions.
- **Python SDKs** (`anthropic` , `openai` libraries) — for reproducible, batched, or pipeline work. Process hundreds of abstracts overnight or build custom tools with structured outputs.
- **Agentic coding tools** — terminal-first agents like **Claude Code** or **OpenAI Codex**, and AI-native editors like **Windsurf** and **Cursor**, can index your entire repository and autonomously execute multi-step tasks across dozens of files. These are a step-change from chat-based coding assistance.

Many research workflows benefit from all three.

Ref: Anthropic docs (docs.anthropic.com); OpenAI docs (platform.openai.com).

1.5 AI Agents

An AI "agent" = an LLM + tools (web search, code execution, file access, APIs) + sequential decision-making.

Current examples: Claude Code, OpenAI Codex, Google Gemini CLI, and deep research modes across all major providers. In astronomy, the vision is agents that could integrate multiple data sources, assess their own confidence, and autonomously trigger follow-up observations on robotic telescopes (Stoppa & Bulmus et al., 2025; NSF-Simons CosmicAI Institute, UT Austin).

> **Rule of thumb:** Treat agents as powerful but unsupervised interns — verify their work before trusting it.

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2. AI "Ethics"

2.1 Data Privacy and Embargoed Data

When you paste text into an LLM, you are sending it to a third-party server. This has direct implications for:

- **Embargoed telescope data** — unreleased JWST, ALMA, or survey collaboration data must never enter a public LLM where it could be used for training
- **Grant proposals and proprietary algorithms** — your competitive advantage could become training data
- Collaborator or student personal information
- Any material under NDA or collaboration agreement (e.g., LSST, DESI data policies)

Key facts:

- Free-tier conversations **may be used for model training**. Paid tiers and API typically offer opt-out, but policies vary.
- A 2025 privacy ranking found that even leading providers offer limited transparency about how user data interacts with training (Incogni, 2025).
- The EDPB published a comprehensive risk assessment of LLM privacy in April 2025, cataloguing risks at every stage of the model lifecycle.

> **Practical rule:** Never paste sensitive or unpublished data into a free-tier LLM. For confidential work, use API access with data retention disabled, or run an open-weight model (Llama 4, DeepSeek) locally on your own machine.

Ref: Incogni, "Gen AI and LLM Data Privacy Ranking 2025"; EDPB, "AI Privacy Risks & Mitigations — LLMs" (April 2025).

2.2 Hallucinations, Bias, and Verification

LLMs fabricate references, invent statistics, and produce plausible-sounding but incorrect scientific claims. This is not a bug — it is inherent to probabilistic text generation.

What the data says:

- A Nature survey of ~3,800 PhD students found **81% do not fully trust AI tools** (Nature, March 2026).
- LLMs overrepresent open-access literature, recent publications, and English-language sources (Wiley, 2025).
- For astronomy, foundational older work and results in non-English journals are frequently underrepresented or absent.

Your obligation: Verify every factual claim, every citation, and every numerical result an LLM produces.

2.3 Transparency and Disclosure

Major publishers and most universities now require disclosure of AI use. The AAS (publishers of ApJ, AJ, ApJS) stipulates that AI programs cannot be listed as co-authors, and you bear 100% responsibility for every error in your manuscript — including those introduced by AI. The expectation:

1. **What** tool you used (model, version)
2. **How** you used it (drafting, analysis, editing, coding)
3. **When** you accessed it
4. That **you take full responsibility** for accuracy

> **Template (from Wiley):** *"The authors acknowledge the use of Claude 3.5 Sonnet (accessed July 2025) to support creation of the abstract and to suggest reorganisation of the Discussion section. All AI-generated suggestions were reviewed, revised, and approved by the authors."*

The cost of non-disclosure is an academic integrity violation. The cost of disclosure is a single sentence in your methods section. Find your university's policy and follow it.

Ref: AAS Editorial Guidelines on Use of AI (2026); Wiley AI Guidelines (2025); Thesify, "Gen AI Policies at the World's Top Universities" (October 2025).

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3. Using AI to Get Work Done (vs. to Shortcut Learning)

3.1 The Learning vs. Productivity Distinction

The single most important principle:

> **AI should amplify your thinking, not replace it.**

The boundary is context-dependent:

✅ Augmenting	❌ Outsourcing
Using AI to explain a concept you're struggling with	Having AI write your assignment
Using AI to debug your code (and learning the fix)	Having AI write your entire pipeline
Using AI to suggest an outline for your chapter	Submitting AI prose as your own
Using AI to check your argument for logical gaps	Accepting AI output without reading it

A useful test: If you could not explain or reproduce the AI's output without the AI, you have outsourced cognition rather than augmented it.

3.2 The AI as Socratic Tutor

One of the highest-value uses of AI for a research student is as an interactive physics tutor — but only if you force it to critique rather than solve. Instead of asking the AI to derive the radiative transfer equation for your cloud geometry, try:

> *"I am trying to derive the radiative transfer equation for this specific cloud geometry. Here is my maths so far. Do not solve it for me, but tell me where my logic or assumptions are failing."*

This "rubber duck debugging" approach — where you articulate your reasoning and the AI probes it — builds understanding in a way that receiving a solution never does. It works equally well for debugging code, checking statistical methods, and stress-testing the logic of a paper draft.

Ref: Harvard GSE AI Policy (2025–2026); Oxford Saïd Business School, "Six Recommendations" (2025); CUNY Academic Works, "AI and Astronomy Lab Manual" (2024/2025).

3.3 In the University and Research Contexts

University policies range from permissive to restrictive, often varying by assignment:

- A 2025 survey found **88% of UK undergraduates** used AI for assessments — up from 53% the year before.
- Top universities (Stanford, Oxford, Harvard) permit AI use with disclosure, while prohibiting it in exams unless explicitly allowed.
- The trend is toward **nuanced, course-specific rules** rather than blanket bans.

As a postgraduate researcher, you occupy a different position: coursework has learning objectives where AI use may be restricted, but your research is professional work where AI is a legitimate tool. **Know the difference, and when in doubt, ask your supervisor.**

In a research group, **AI use should be transparent to your collaborators**. If you used an LLM to draft a paper section, your co-authors need to know — they are signing off on the content. In large collaborations (LSST, DESI, SKA), AI-generated code that you don't fully understand won't survive internal review, and shouldn't.

> **Good practice (from Oxford Saïd):** Prompt the AI to *challenge* your ideas rather than affirm them. Say "no" to its first output three times — this forces more nuanced results and prevents intellectual complacency from accepting flattering AI output.

Ref: Thesify (October 2025); Stanford Academic Integrity Working Group (2025); Oxford Saïd Business School (2025); IHS (October 2025).

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4. Smart Prompting

4.1 Core Techniques That Actually Work

Research across 1,500+ papers and 200+ techniques (Schulhoff et al., co-authored with OpenAI, Microsoft, Google, Princeton, Stanford) has converged on what reliably improves output:

Technique	Impact	How
Few-shot prompting	Highest impact — 0% → 90% in documented cases	Provide 2–5 examples of the input-output pattern you want
Chain-of-thought	Major improvement on multi-step problems	Ask the model to reason step-by-step
Providing context	Consistently underrated, massively impactful	Include background info, constraints, relevant data
Specifying output format	Prevents vague, unusable responses	"Return a table with columns: title, author, year, finding"
Role prompting	⚠️ Minimal effect on correctness	May influence tone/style, but don't rely on it for accuracy

4.2 Structuring Complex Prompts

For non-trivial tasks, treat your prompt like a specification document. Include:

1. **Task description** — what you want done
2. **Context/constraints** — domain knowledge, limitations, audience
3. **Output format** — table, bullet list, code, paragraph, JSON
4. **Examples** — even one good example dramatically improves output

Use clear delimiters (Markdown headers, XML tags, numbered sections) to separate components. Research shows formatting differences alone can cause accuracy variations of **up to 76 percentage points** (Getmaxim.ai, 2025).

> **Bad prompt:** *"Summarise these papers."*
>
> **Good prompt:** *"I need to extract stellar parameters from the following text and format them into a LaTeX* `aastex631` *deluxetable suitable for ApJ. Here is exactly how one row should look: [example row]. Now process the rest of the text."*

4.3 Chain-of-Thought for Physics Workflows

For code generation tasks with physical assumptions, force the AI to show its reasoning before writing code:

> *"Before writing the Python script to extract transit light curves from this FITS file, outline: (1) the physical assumptions you're making, (2) the background subtraction method, and (3) the mathematical steps. Then I'll review the plan before you write code."*

This lets you catch physics errors in plain English before they become buried in Python. It also creates a record of the assumptions, which is valuable for your methods section.

4.4 Iteration and Prompt Libraries

The best practitioners treat prompting as an experimental process:

- **Build, test, refine.** When output is poor, diagnose: insufficient context? Ambiguous instructions? Wrong model?
- **Save what works.** Keep a prompt library (a text file, Obsidian vault, or git repo).
- **Decompose complex tasks.** Break into subtasks and prompt each separately — e.g., first extract methods, then compare across papers, then synthesise.
- **Use self-criticism.** Follow up with: *"Now review your output for errors, omissions, and unsupported claims."*

Ref: Schulhoff et al.; Sahoo et al., arXiv:2402.07927 (2024/2025); Promptingguide.ai; Lakera (2026).

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5. AI to Communicate

5.1 Emails, Notes, and Summaries

AI excels at first drafts of routine communication. The workflow:

1. Provide the **key points**, **audience**, **tone**, and **constraints**
2. Generate a draft
3. **Edit it to sound like you** — this step is non-negotiable

This is particularly valuable for non-native English speakers — AI can polish grammar and idiom without flattening your voice. Tools like **Writefull** are specifically designed for academic English.

> **Warning:** Sending unreviewed AI text is both obvious to recipients and a reputational risk. Always read before sending.

Ref: IHS (October 2025); Avidnote (2025).

5.2 Reports, Papers, and Theses

For longer-form writing, AI is best as a **structural assistant**, not a ghostwriter:

- **Generate outlines** and suggest paragraph transitions
- **Identify gaps** in your argument
- **Act as a critical reviewer** — prompt it to find holes in your reasoning before submission
- **Handle mechanical tasks** — formatting references, checking consistent terminology, verifying every figure is cited

Beating blank-page syndrome: If you're staring at an empty LaTeX document, dump your messy lab notes, bullet points, and plot descriptions into an LLM and ask it to draft a rough Methods section. The output will be dry and generic, but editing a mediocre draft is psychologically far easier than starting from nothing. You maintain your scientific voice by editing heavily — the AI provides scaffolding, not content.

For theses: your examiners are looking for evidence of *your* critical thinking. AI-generated prose, however polished, lacks the specificity that comes from deep engagement with your data.

Ref: Oxford Saïd (2025); Wiley AI Guidelines (2025).

5.3 Reverse-Engineering Proposals and Talks

For telescope time proposals and grant applications, a powerful technique: feed a successful, funded proposal (from your supervisor or a public archive) into an LLM with a large context window. Ask it to extract the structural template, the rhetorical strategies used to justify the science case, and the tone. Then use that extracted template as the skeleton for your own original science case. This teaches you what a good proposal looks like while ensuring you write your own.

For conference talks, a similar workflow:

1. Give the AI your paper abstract and key figures
2. Ask it to suggest a 12-minute talk structure
3. Iterate on the narrative arc
4. Build the slides yourself

AI can also help **translate between audiences** — taking a technical result and reframing it for a departmental colloquium vs. a public lecture vs. a grant committee.

Ref: Nature (March 2026); SciPy 2025, Physics & Astronomy track.

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6. AI for Navigating the Literature

6.1 Discovery: Finding What Exists

Traditional keyword searches miss semantically related work. AI-powered tools fill this gap:

Tool	What it does	Good for
Elicit	Searches academic databases with natural-language questions, synthesises across papers	Evidence gathering, systematic reviews
Research Rabbit	Builds citation network visualisations from seed papers	Mapping a field, finding related work
Connected Papers / Litmaps	Visual maps of research fields from a single seed paper	Understanding how ideas evolved
Scite	Shows how papers are cited downstream — supporting, contrasting, or mentioning	Assessing whether a result has held up
Consensus	Evidence-based answers with linked citations	Quick factual questions with scholarly backing

⚠️ **Important caveat:** Many of these tools are built on Semantic Scholar, which is extensive but smaller than Google Scholar or ADS. Use them to *complement*, not replace, your standard database searches.

Ref: Oklahoma State University Library (2025); Nature (March 2026).

6.2 Large Context Windows as Data Extraction Engines

Modern models accept 200K–1M+ tokens of context. This enables a workflow that was impossible two years ago: drop 30–50 peer-reviewed PDFs directly into a long-context model and prompt:

> *"Read these papers and extract into a CSV: the reported exoplanet radii, their error margins, the spectrograph used, and the reference for each measurement."*

This grounds the AI entirely in the literature you provide (eliminating hallucination risk for the source material) and turns it into a structured data-extraction engine. The same technique works for extracting stellar parameters, survey completeness limits, or methodology comparisons across a body of work.

> **Critical caveat:** Extraction accuracy is not 100%. Always spot-check a sample of rows against the original papers, especially for numerical values.

Ref: Scite.ai; Berkeley/Polymathic AI, Multimodal Universe dataset (NeurIPS 2024); Avidnote (2025).

6.3 Systematic Reviews and Synthesis

For rigorous literature work, use dedicated tools rather than general-purpose LLMs:

- **Rayyan** — AI-assisted screening for systematic reviews
- **Elicit** — structured data extraction across hundreds of papers
- **Deep research features** (ChatGPT, Gemini, Claude, Perplexity) — useful for initial field mapping, but citations need manual verification and paywalled content is often missed

> **Astronomy-specific:** The Multimodal Universe dataset (100 TB, released via Hugging Face) brings together observations from DESI, SDSS, and other surveys in ML-ready format. Tools built on it will increasingly support literature-scale and data-scale queries.

Ref: Aaron Tay, "The Rise of Agent-Based Deep Research" (2025); Wiley AI Guidelines (2025).

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7. AI Coding

7.1 AI as a Coding Partner

AI coding assistants (GitHub Copilot, Claude Code, agentic IDEs like Windsurf and Cursor) can dramatically accelerate development. But the research paints a nuanced picture:

What works well:

- Generating boilerplate and scaffolding
- Suggesting function implementations for well-understood patterns
- Writing unit tests
- Explaining unfamiliar codebases
- Translating between languages and modernising legacy code

What doesn't (yet):

- Complex refactoring — LLMs struggle with maintaining behaviour while restructuring (Argonne National Laboratory, 2025)
- Unsupervised operation — they introduce subtle bugs that pass casual inspection
- Deep knowledge transfer — a Saarland University study found developers using Copilot accept suggestions with minimal scrutiny, reducing learning

> **The rule:** Use AI to write code faster, but **always read and test** what it produces. In large collaborations (LSST, DESI, SKA), "vibe-coded" pipelines you don't fully understand won't survive internal review — and shouldn't.

Ref: Dubey & Dhruv, Argonne National Laboratory (July 2025); Saarland University study, The Register (November 2025).

7.2 Legacy Code and Shell Scripting

Much of astronomy runs on legacy code. A high-value use of AI is **translating old Fortran 77 or IDL routines** (inherited from your supervisor or the literature) into modern, documented Python using `astropy`, `numpy`, or `jax`. Agentic tools that can index an entire codebase (Claude Code, Windsurf, Cursor) handle this far better than chat interfaces, because they can see the dependencies and call structure.

Shell scripting is the glue of data reduction. AI is exceptionally good at generating:

- FITS header parsing across directory trees
- SLURM/PBS job submission scripts for HPC clusters
- Batch file renaming, format conversion, and piping between reduction tools

> **Prompting tip:** Be specific about your environment — *"I'm on macOS using zsh, data is in FITS format across 200 directories named by observation date, and I need to extract EXPTIME and FILTER from each header into a CSV."*

Ref: ASPP School (aspp.school); SciPy 2025.

7.3 Write the Physics, Let AI Write the Tests

Python dominates astronomy (`astropy` , `numpy` , `scipy` , `matplotlib`), and AI assistants are highly competent with the entire scientific Python stack. But the sharpest workflow isn't "AI writes code, you review it" — it's a division of labour:

- **You** write the core mathematical and physical functions (so you understand and own the science)
- **AI** writes the documentation, boilerplate, and — critically — the test suite

> *"Write a comprehensive `pytest` suite for this function, including edge cases for negative fluxes, NaN values, and empty arrays."*

This is test-driven development applied to scientific code: AI is mediocre at inventing novel physics but excellent at generating systematic test coverage for the code you've written.

Critical practices remain:

- **Always run and test** AI-generated code — never submit untested code to a cluster
- **Be especially cautious with numerical code** — silent precision errors can corrupt results
- **Understand what the code does** — if you can't explain it, you can't debug it at 2am during your observing run

Ref: MIT News (April 2025); Argonne National Laboratory (2025); ASPP School.

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8. AI for Project Management

8.1 Task Tracking and Research Organisation

A PhD involves hundreds of tasks across multiple timescales — observations, reductions, analysis, writing, meetings, deadlines. AI can help manage this.

Simple, high-value approach: At the end of each week, paste your notes into an LLM and ask it to generate:

- A structured summary of progress
- Outstanding tasks, prioritised
- Blockers and dependencies
- Questions for your next supervisor meeting

This creates a running log invaluable for supervisor meetings and thesis writing. Cost: five minutes per week.

Breaking down the overwhelm: "Write Chapter 3" is paralysing. Feed your thesis chapter outline to an LLM and ask it to decompose it into concrete 2-hour tasks: *"Draft 3 paragraphs on the history of 21cm cosmology instrumentation"* is something you can finish before lunch.

Ref: PPM Express (2025); Nature (March 2026); Bail et al., "LLMs as Research Tools," arXiv (2024/2025).

8.2 Meeting Notes and Supervisor Communication

- **Otter.ai** or built-in Zoom/Teams transcription can record supervisor meetings → feed the transcript to an LLM → get structured action items.
- For **journal clubs**, AI can generate structured summaries of the paper being discussed, freeing you to focus on the actual discussion.
- For **cross-timezone collaboration**, an LLM can draft concise email summaries of decisions for collaborators who missed a meeting.

This solves the perennial problem of leaving a meeting with a vague sense of what was agreed but no written record.

Ref: IHS (October 2025); Get Harvest (2025).

8.3 Automated Knowledge Management

Rather than manually scanning arXiv each morning and forgetting what you read, build an automated pipeline:

1. Set up **arXiv RSS feeds** filtered to your sub-field (e.g., `astro-ph.SR`, `astro-ph.EP`)
2. Use a lightweight LLM API (e.g., GPT-5.4 mini, Claude Haiku) to summarise each day's abstracts and score relevance to your thesis topic
3. Push the results into an **Obsidian vault or Notion database**, auto-tagged by sub-topic

This can be done with a short Python script and a cron job, or with a no-code tool like Zapier. The result: a searchable, pre-filtered literature feed that builds itself while you sleep.

8.4 Time and Scope Management

Research projects are notorious for scope creep. AI can help by:

- **Structuring a project plan** from your research question — phases, dependencies, parallel work
- **Estimating realistic timelines** given constraints (telescope time, computing resources, collaborator availability)
- **Identifying the critical path** — what must happen sequentially vs. what can overlap
- **Protecting deep-work time** — tools like Motion AI auto-schedule focused blocks around existing commitments

> This is not a replacement for your supervisor's experience. It's a structured starting point for the conversation.

Ref: PMI, "AI in Project Management Global Report" (2025); Avidnote (2025).

Discussion Points for Class

1. **Where is the line** between using AI to learn and using AI to bypass learning? Does it shift between coursework and research?
2. **What would you never paste** into ChatGPT? Where is your personal data-sensitivity boundary? What about embargoed survey data?
3. **Try it now:** Take a concept from your research area. Prompt two different LLMs. Compare the outputs. What do you notice?
4. **Astronomy-specific:** Vera Rubin's LSST will generate 10 million alerts per night from 2026. How does this change the role of AI in your field?
5. **Your institution's policy:** Have you read it? Can you find it? What does it actually say?
6. **The Socratic test:** Can you prompt an AI to critique your reasoning on a current problem *without* solving it for you? Try it.
